

Lab 6

System Measurements and Lab Equipment

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General Objectives

The objectives of this lab include familiarizing ourselves more with the lab equipment such as the oscilloscopes and function generator, understand how to verify the functionality of components and verify that components meet the requirements we have for them in preparation for when we assemble our final project.

Procedure

Equipment and supplies

- Oscilloscope
- Oscilloscope Probes
- Function Generator
- Function Generator Clamps (BNC-to-alligator)
- Jumper Wires
- Lab Components from Provided Lab bench Box

Procedure

In this lab, the first thing we did was verify that we had our necessary components in our kit so we can complete the objectives of this session. We then reviewed the lab instructions to make sure we were properly prepared for the elements of this lab session. We then analyzed each daughter-board from the stereo amplifier; specifically analyzing only the left channel of each board. We began with the line buffer and its performance. The results of this are found in figures 1 and 2. We analyzed the gain of the audio amplifier in this part of the instructions. After recording our necessary information from both the input and output of the circuit using the oscilloscope and comparing to our simulated values from earlier in the semester. Using information recorded in table 1, we also were able to verify the behavior of the circuit and if it matched what we were expecting. Below table 1 will be found the maximum deviation of the output voltage over frequency.

Next, we analyzed both the Microphone Preamp circuit and the Audio Mixer and the behavior of each one. Using the oscilloscope, the input and output was analyzed for our two frequencies again. This is shown in figures 3, 4, and 5 for these circuits with table 2 recording values we collected. We dropped the voltages however for these circuits so we didn't saturate the opamp in the circuit. Here we again only analyzed the left channel in regards to the Mixer.

We analyzed the Treble and Bass Circuit performances in the same manner we have with all of the other daughter-boards. Here we analyzed only the left channel of each circuit; we recorded our collected values in table 3 and below are also images of our oscilloscope readings in figures 6 through 9 for the treble readings and figures 10 through 13 for the bass readings.

After ending the lab, we then made sure everything was properly stowed away as indicated in the final images of this report. We found that these circuits do indeed fit the simulations that we had performed previously in prior lab sessions for each given circuit and board.

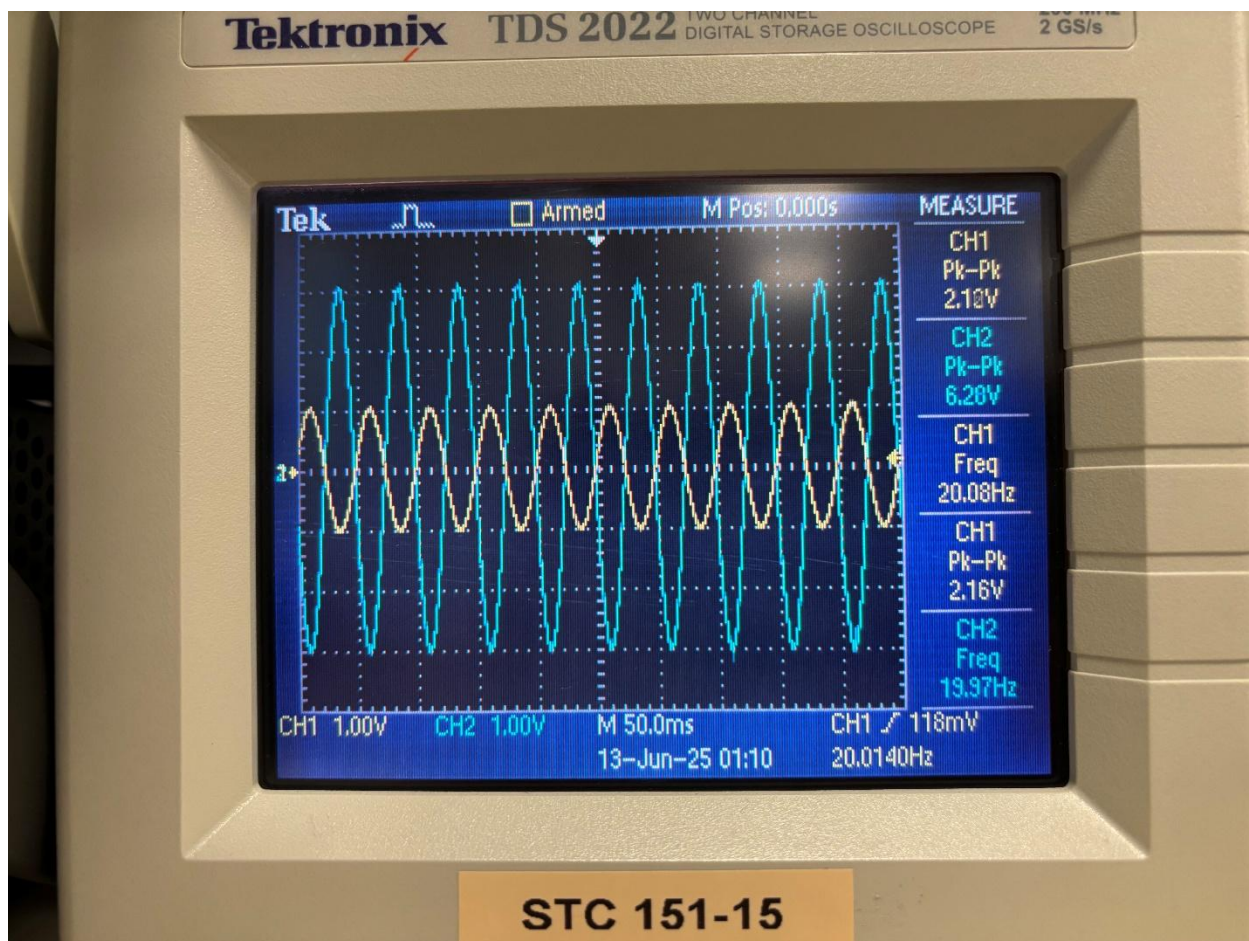


Figure 1 Frequency Response 20 Hz

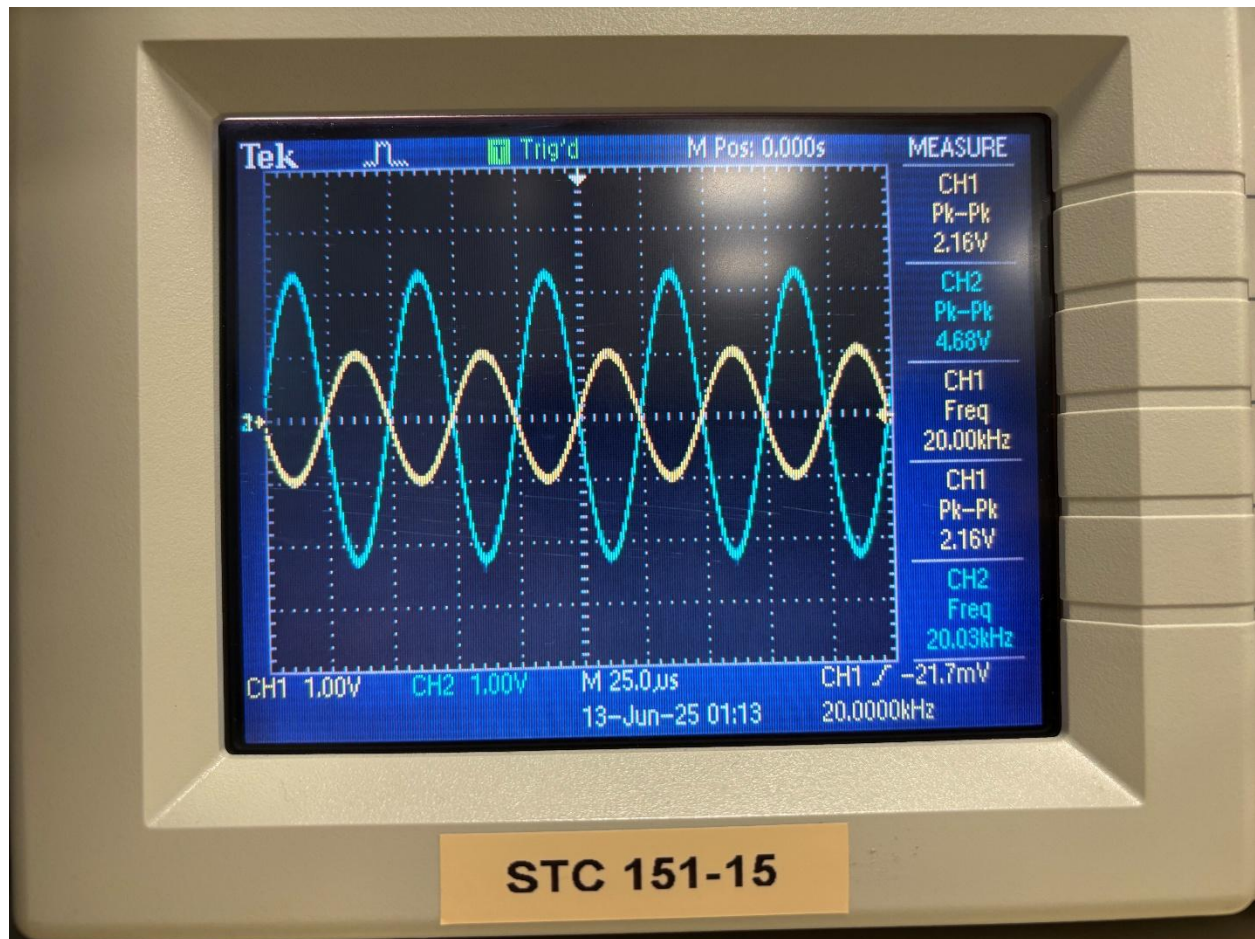


Figure 2 Frequency Response 20 kHz

Table 1 Frequency Response.

$V_{in} = 2V_{pp}$ from the function generator	Frequency	$V_{in(pp)}$	$V_{out(pp)}$	Gain: $A = \frac{V_{out(pp)}}{V_{in(pp)}}$
Left Channel	20 Hz	2.16V	6.12V	2.83V
Left Channel	20 kHz	2.16V	4.68V	2.17V

Calculate the maximum deviation of the output voltage:

$$\text{Deviation}_{\text{Right}}(\text{dB}) = 20 \times \log_{10} (A_{R(\text{max})}/A_{R(\text{min})}) = 20 \times \log_{10} (2.83/2.17)$$

$$\text{Deviation}_{\text{Right}}(\text{dB}) = 2.31\text{dB}$$

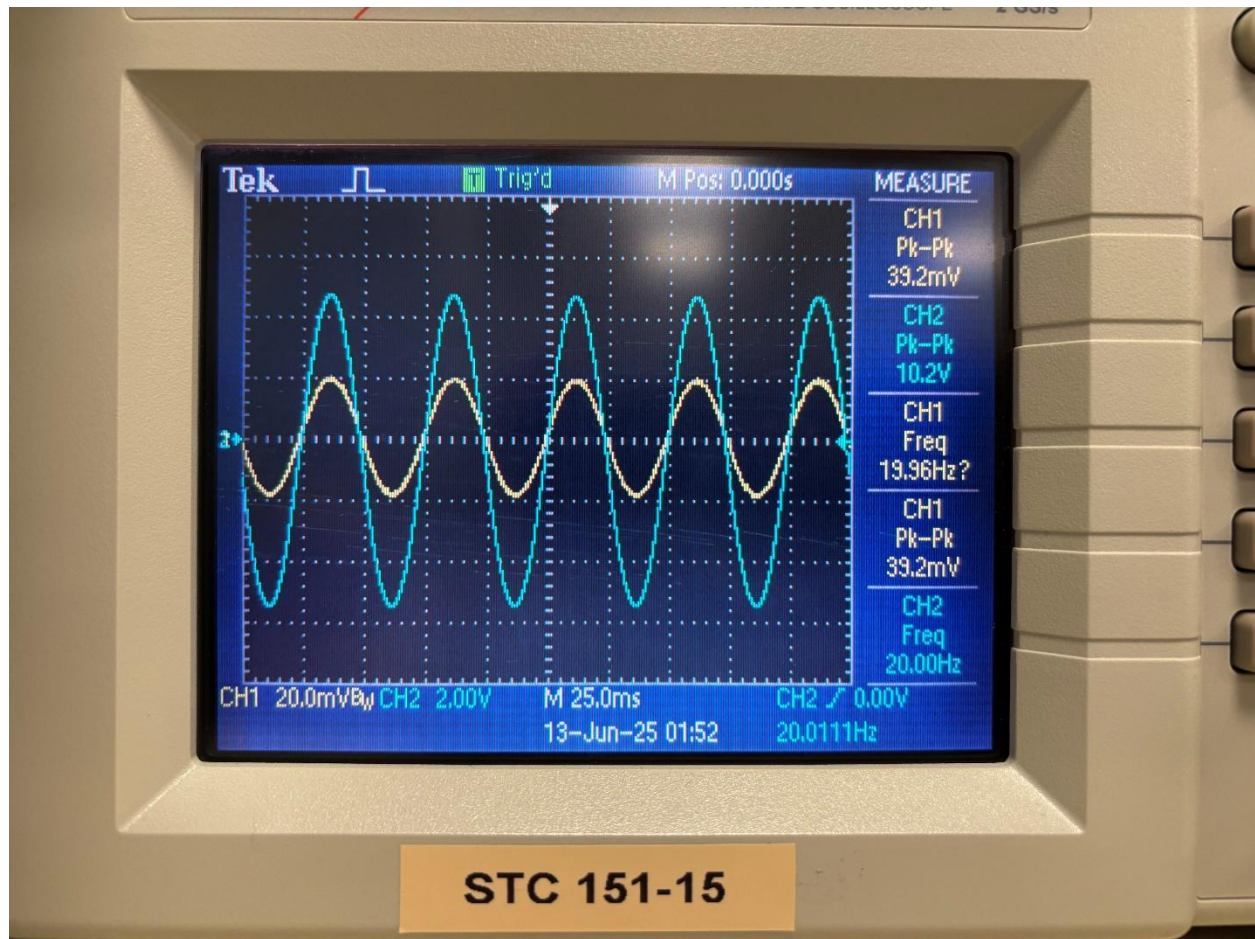


Figure 3 Pre-Amp Frequency Response 20 Hz

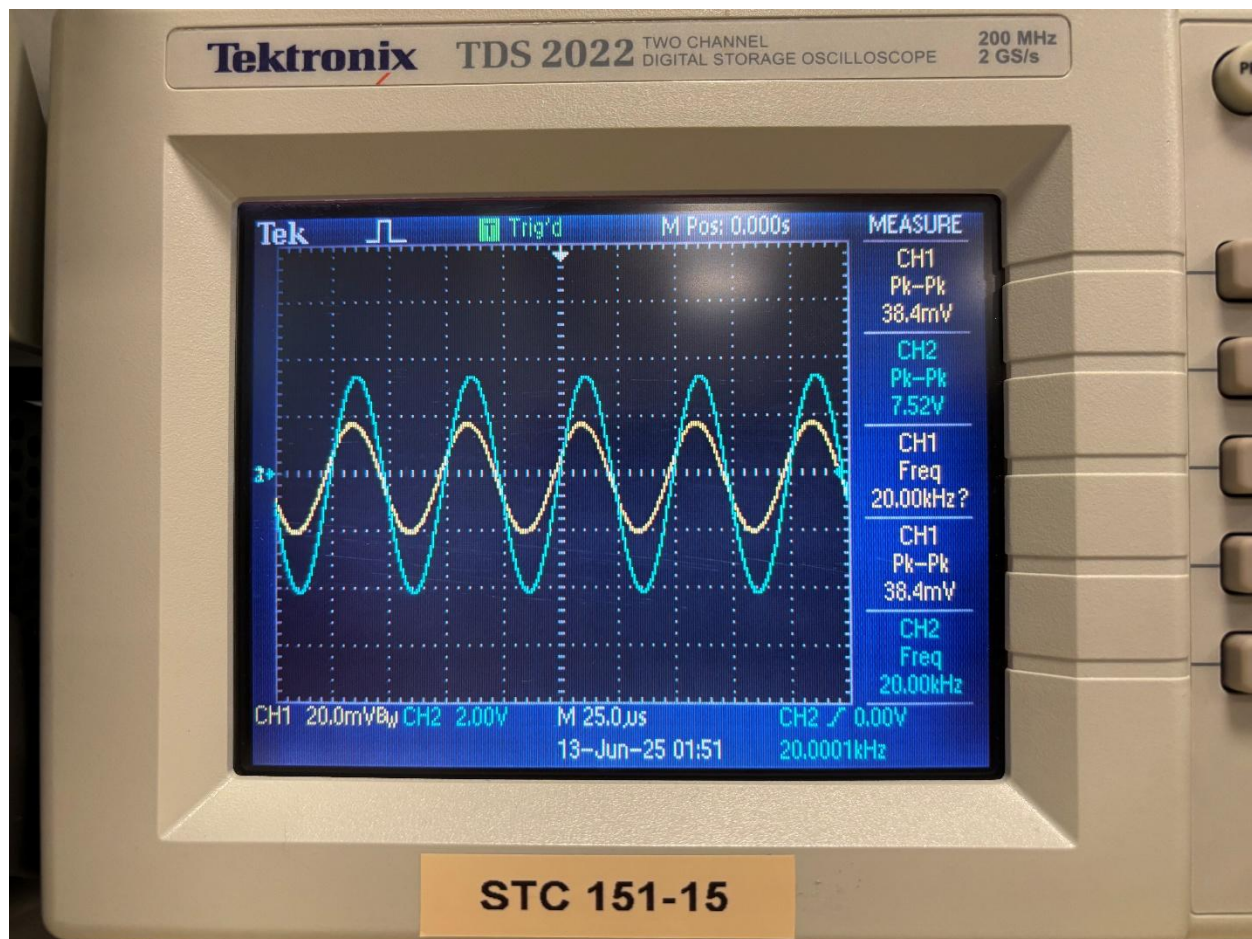


Figure 4 Pre-Amp Frequency Response 20 kHz

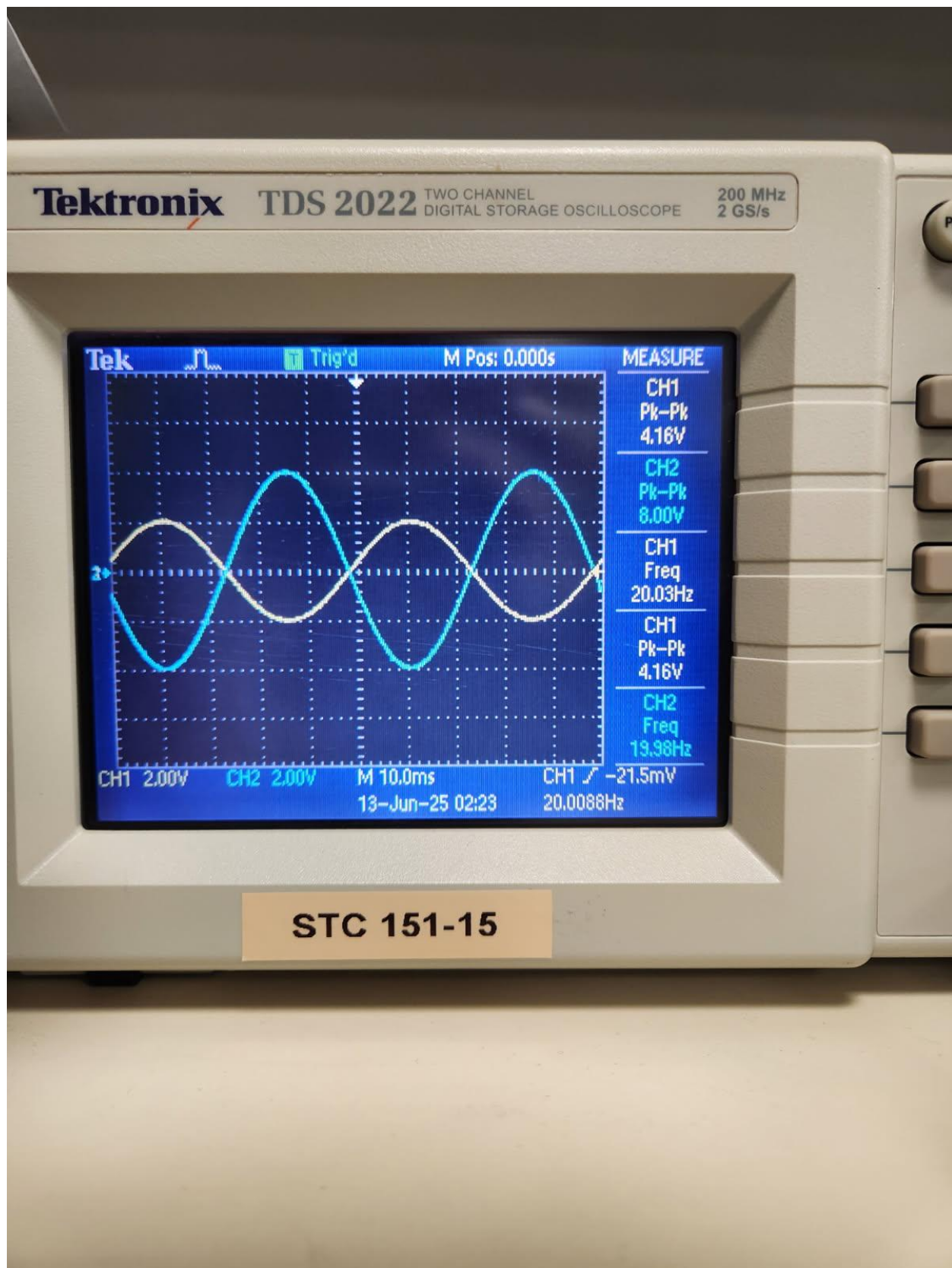


Figure 5 Mixer Frequency Response 20 Hz

(Mixer Frequency Response 20 Hz and 20 kHz graphs and $|A_{dB}|$ are approximately equivalent)

Table 2 Pre-Amp and Mixer Frequency Response.

Amplifier	Frequency	V_{in}	V_{out}	$ A_{dB} $
Pre-Amp	20 Hz	39.2mV	10.2V	48.3dB
	20 kHz	39.2mV	7.5V	45.6dB
Mixer Left Channel	20 Hz	4V	8V	6.02dB
	20 kHz	4V	7.92V	5.92dB

Treble Frequency Response Measurement Figures.

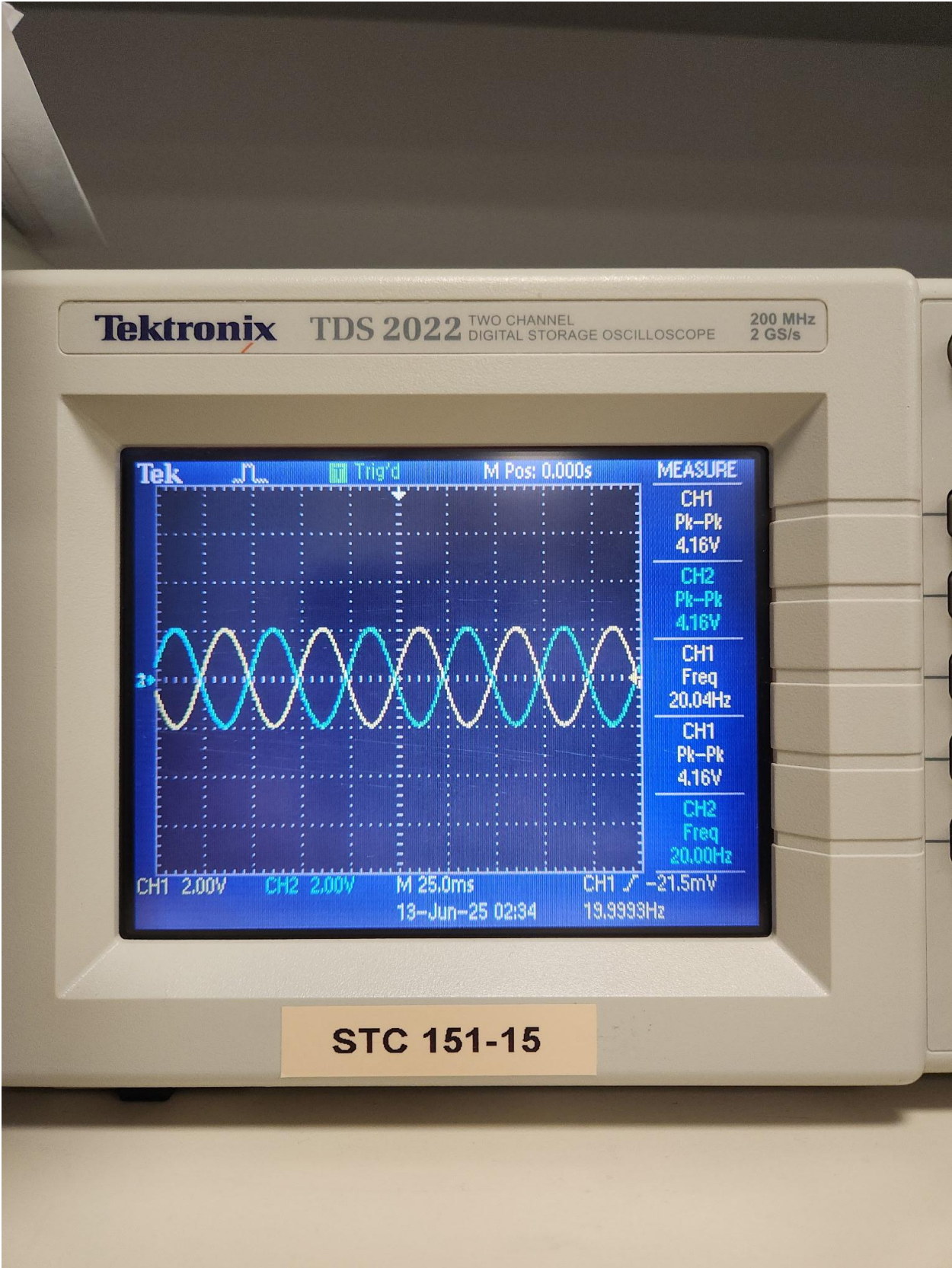


Figure 6 (a = 0) Treble 20 Hz

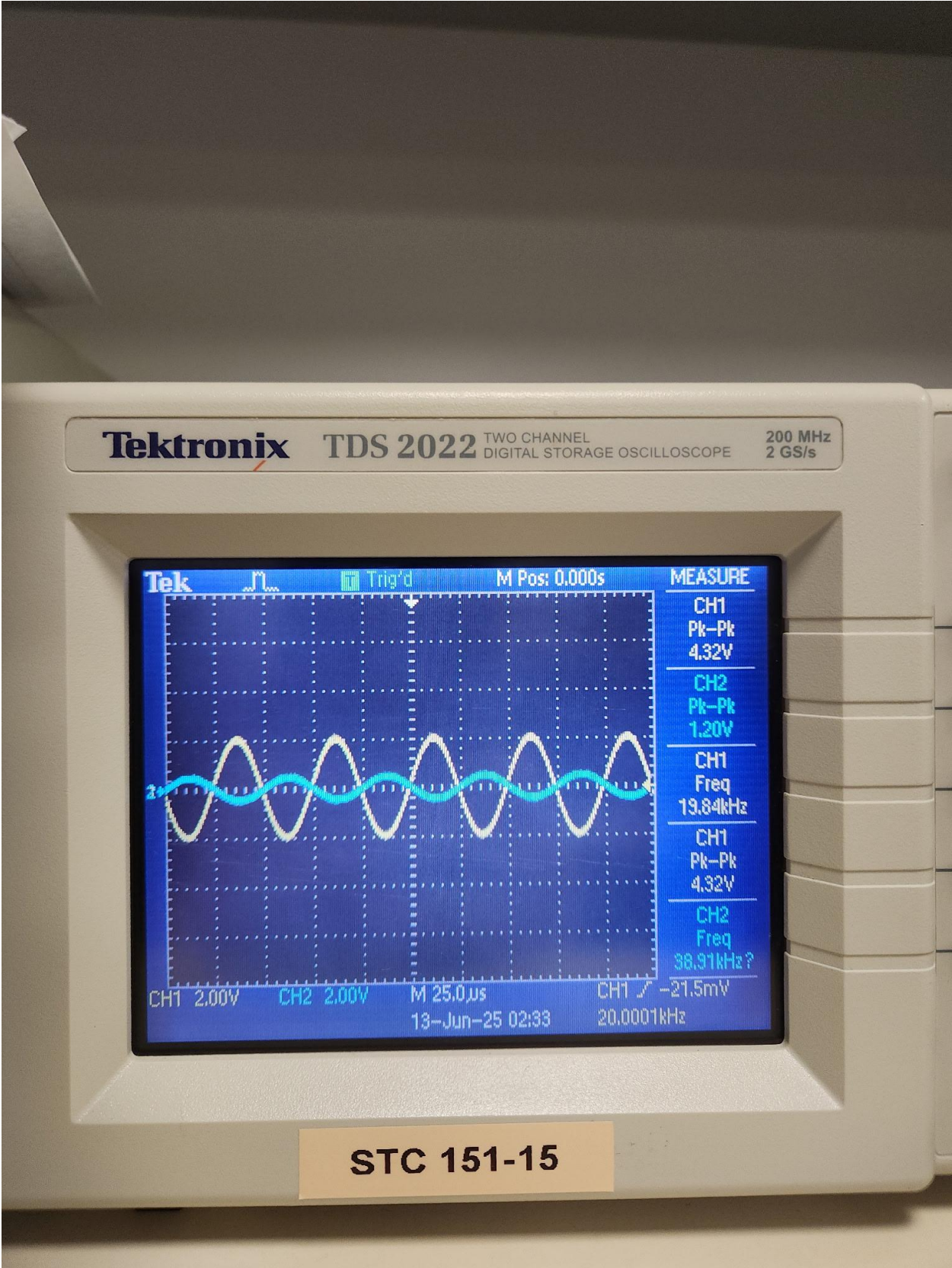


Figure 7 ($a = 0$) Treble 20 kHz

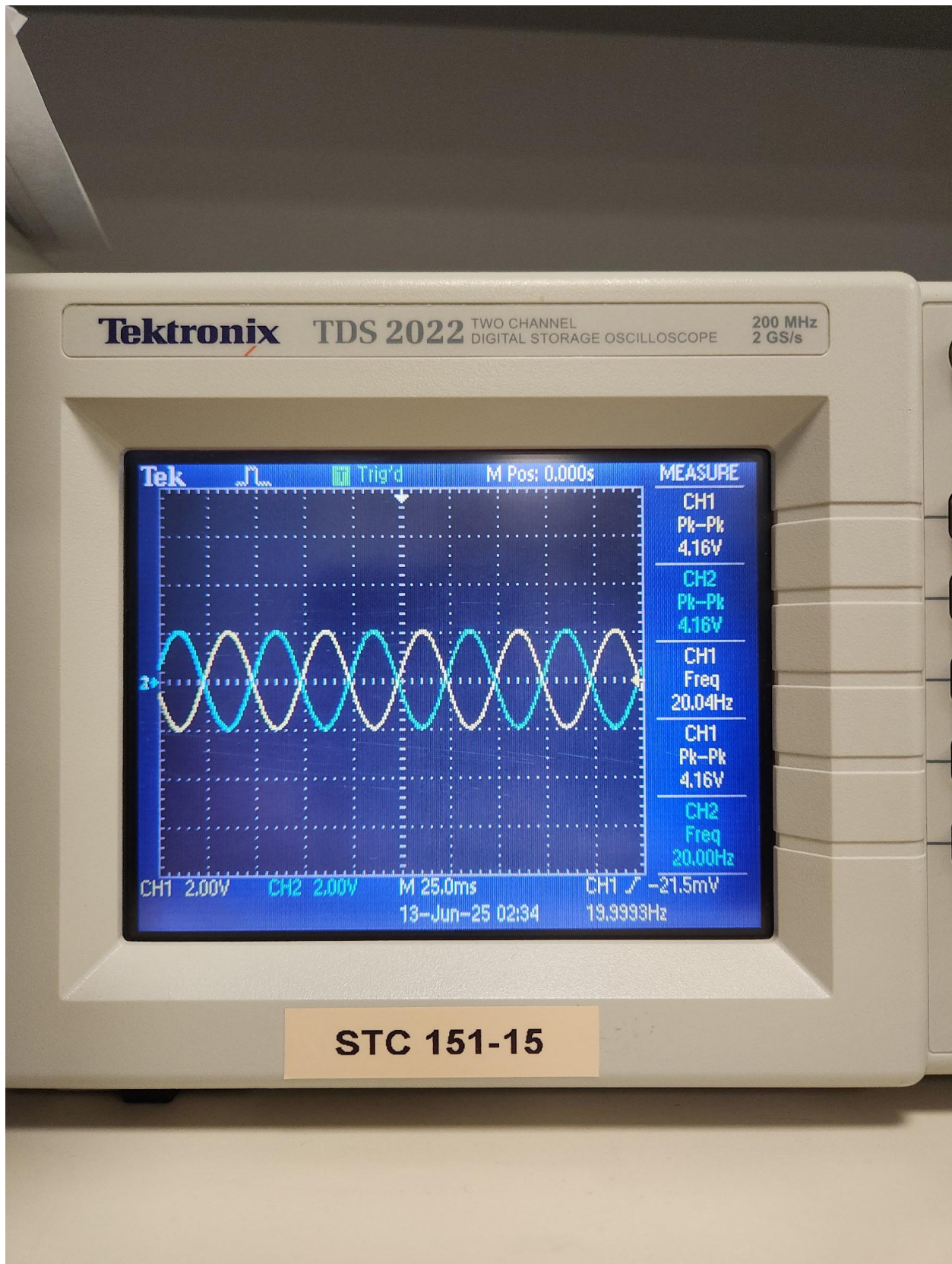


Figure 8 (a = 1) Treble 20 Hz

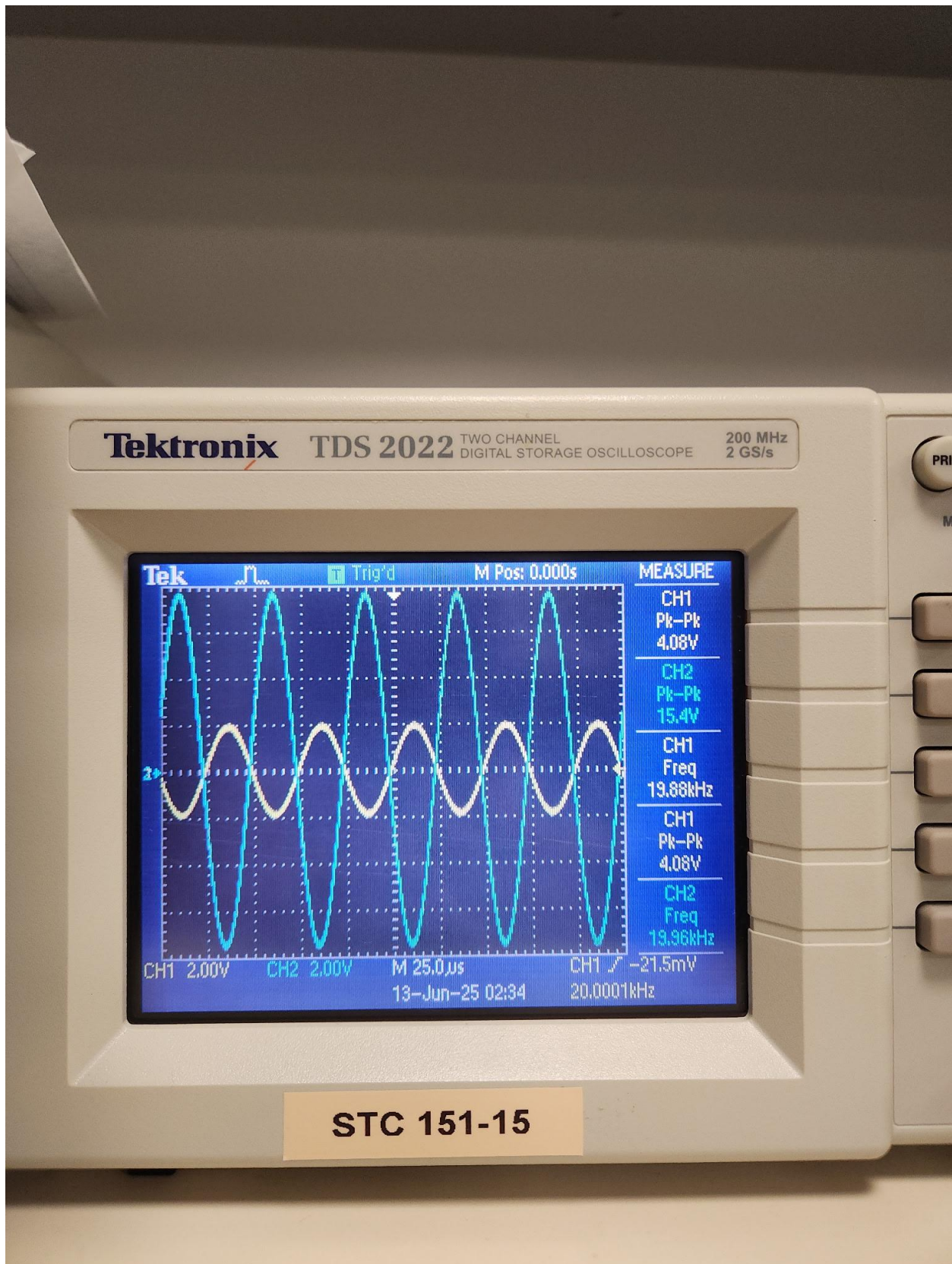


Figure 9 (a = 1) Treble 20 kHz

Bass Frequency Response Measurement Figures.

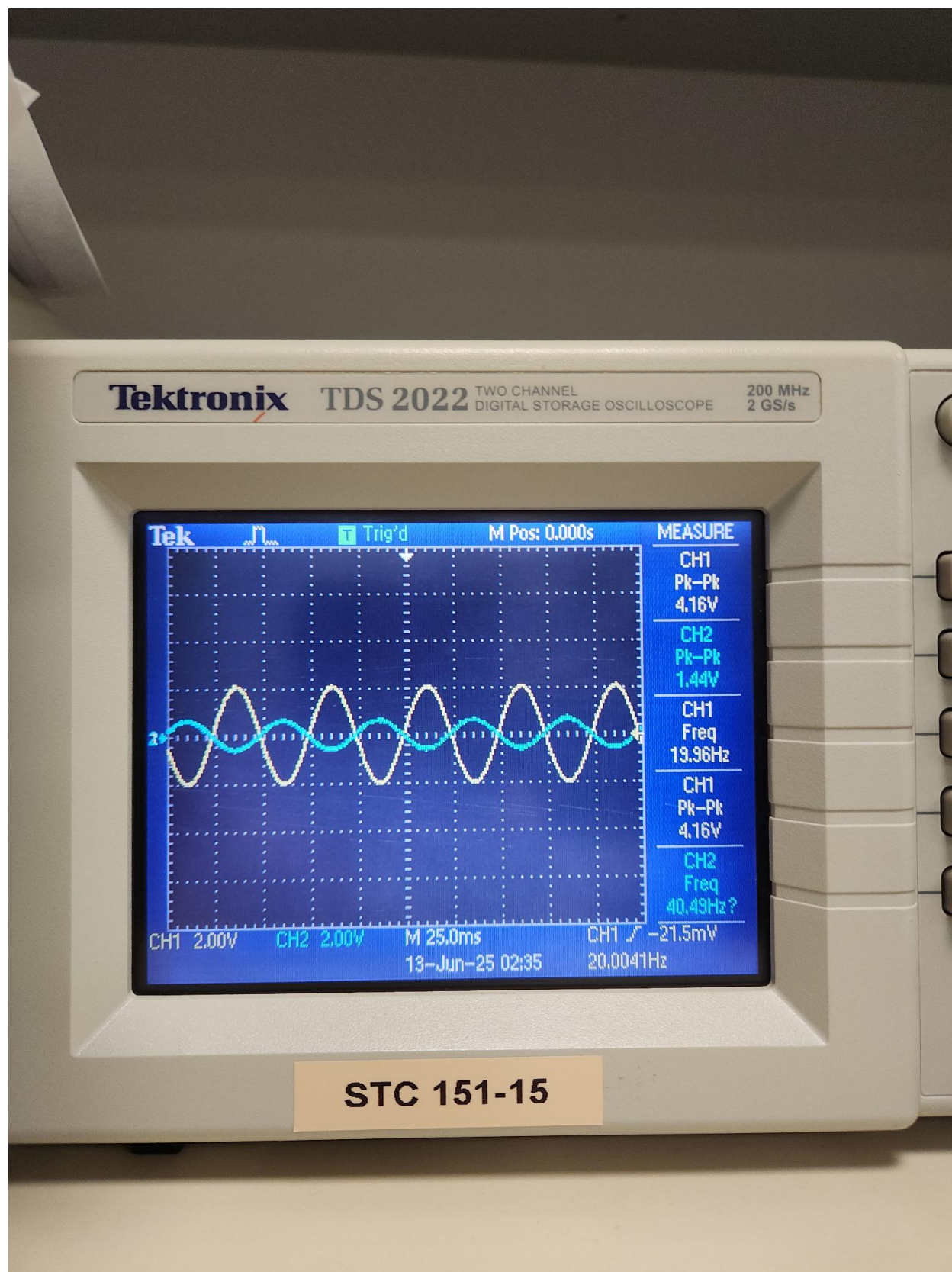


Figure 10 ($a = 0$) Bass 20 Hz

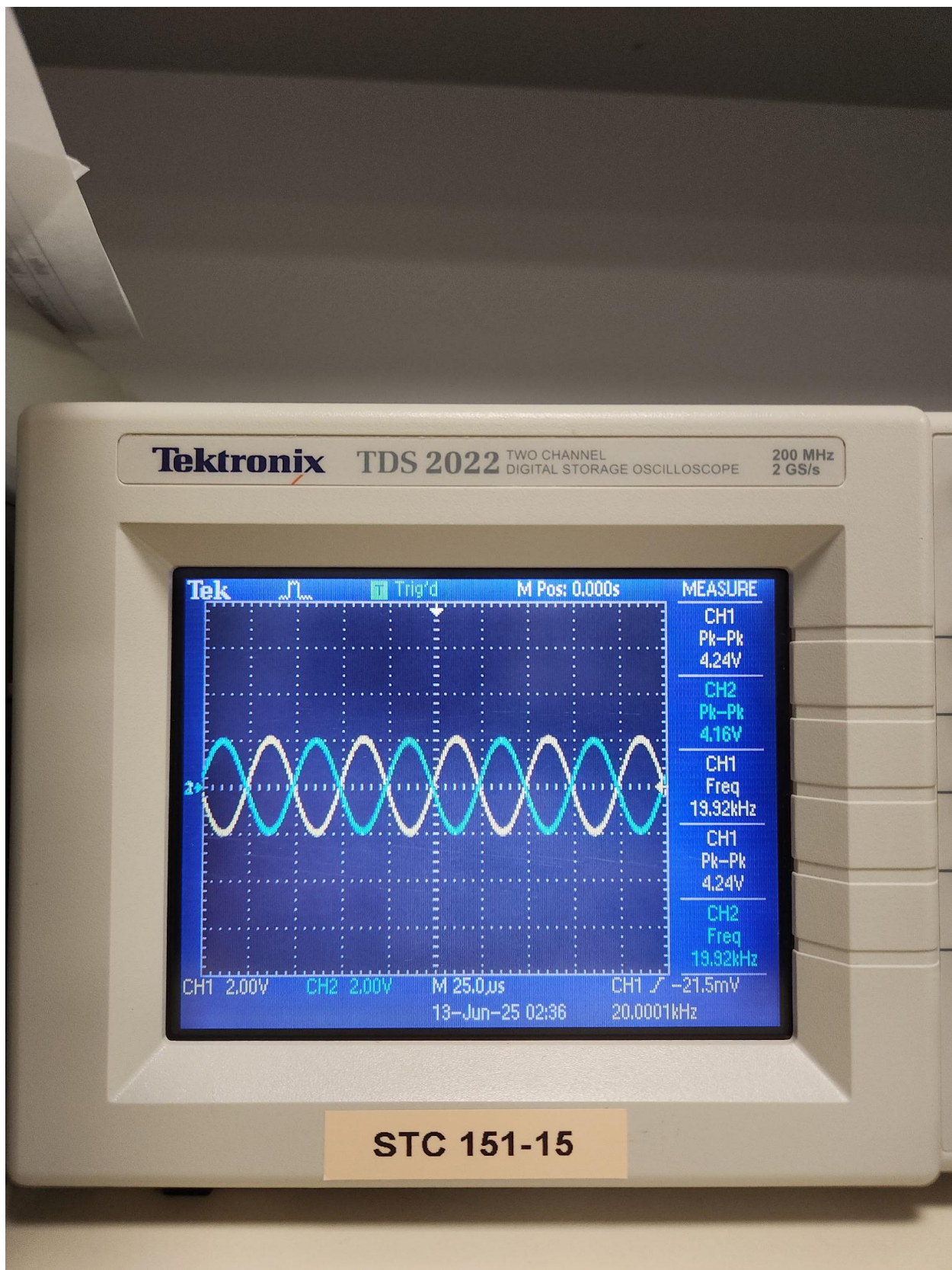


Figure 11 ($a = 0$) Bass 20 kHz

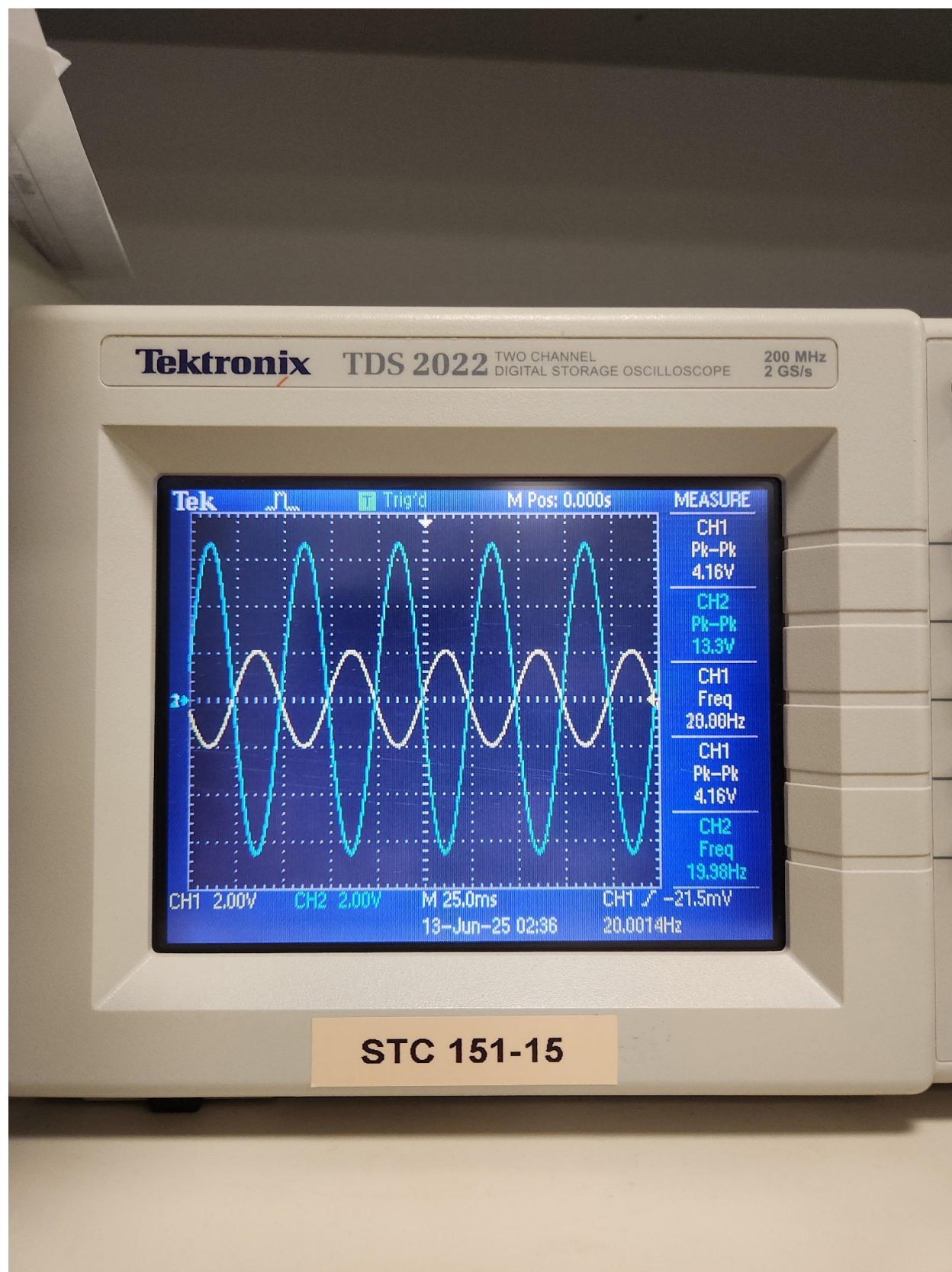


Figure 12 (a = 1) Bass 20 Hz

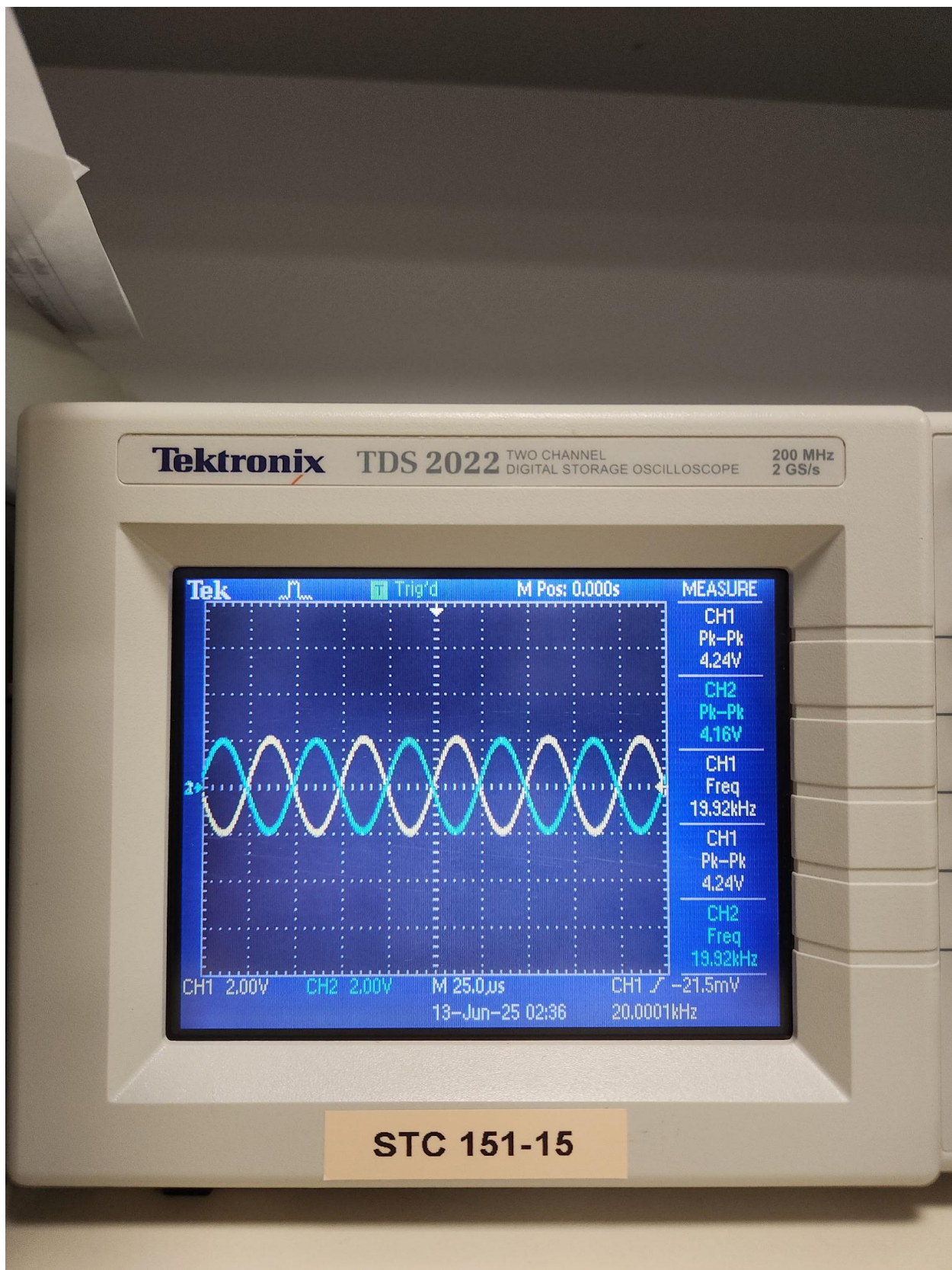


Figure 13 ($a = 1$) Bass 20 kHz

Table 3 Treble and Bass Frequency Response Measurements.

	Channel	$f(\text{Hz})$	Treble			Bass		
			V_{in}	V_{out}	$ A_{\text{dB}} $	V_{in}	V_{out}	$ A_{\text{dB}} $
$a = 0$	Left	20 Hz	4.24V	4.24V	0dB	4.24V	1.44V	-9.38dB
		20 kHz	4.24V	1.2V	-10.96dB	4.24V	4.16V	-0.16dB
$a = 1$	Left	20 Hz	4.24V	4.24V	0dB	4.24V	13.3V	9.92dB
		20 kHz	4.24V	15.4V	11.2dB	4.24V	4.16V	-0.16dB

Laboratory Cleanliness Figures.



Figure 10 Station Drawer Organization



Figure 12 Lab Bench Organization

Conclusion

In this lab, we did meet all of the objectives of our lab and successfully checked each daughterboard, and the motherboard in the process. One of the issues we had that caused us to run very late with this lab session was our measurements were not coming into the oscilloscope as we were expecting and this was causing some issues. We spent a significant amount of time with one of the TA's to try and troubleshoot this and ended up discovering that one of our oscilloscope probes were not properly calibrated; this was the reason we were having issues very quickly into our lab session. The TA ended up fixing this while we used a different probe, and we recorded the data we measured and found that it was now giving accurate data to us. This is in fact one of the ways we became more familiar with the oscilloscope, having to troubleshoot some of the settings on it and then learning of the possibility of the probe being inaccurate gave more insight in what we can keep in mind when we find our data is not what we expect from a given circuit. Throughout this lab, we were able to verify that the boards met our expectations and fit the design requirements in their behavior.